

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-1

DURATION: 25 MINUTES

WINTER SEMESTER, 2024-2025

FULL MARKS: 15

CSE 4513: Software Engineering and Object-Oriented Design

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer all 4 (four) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. Who is primarily responsible for requirements gathering in a software development project? Briefly explain. 3
(CO1)
(PO1)

Solution: Business Analysts are the professionals who efficiently carry out software requirement gathering by breaking down the critical technical specifications into effective documentation and user stories.

2. Describe the relationship between SRS and design documents in the Design phase of SDLC. 6
(CO1)
(PO1)

Solution: In the Design phase of the Software Development Life Cycle (SDLC), design documents are prepared based on the Software Requirements Specification (SRS). The SRS defines *what* the system is required to do by specifying functional and non-functional requirements. Design documents translate these requirements into *how* the system will be implemented. The SRS serves as the primary input to the design phase. Using the SRS, designers develop the system architecture, module structure, database design, and interface specifications.

High-Level Design (HLD) maps the requirements in the SRS to major system components and their interactions, while **Low-Level Design (LLD)** provides detailed logic, data structures, and inputs and outputs for each module.

Therefore, design documents must strictly conform to the SRS to ensure that all specified requirements are correctly and completely implemented.

3. *Weird Things* is an upcoming popular science-fiction series on **Banglaflix**, following a group of kids uncovering mysterious and unexpected events in their town. The story starts simply but gradually introduces complex twists involving powerful forces and uncertain outcomes. 6
(CO2)
(PO2)

A software team is developing a *Weird Things*-based application. At the start, the team follows a strict step-by-step development approach with fixed requirements. As the story unfolds, unexpected twists involving Pecna and uncertainty about Twelve's fate introduce new requirements and risks. To manage these changes, the team shifts to a development approach that works in repeated cycles with regular reviews.

1. Identify the SDLC model used in the initial phase that follows the sequential approach.
2. Which SDLC model is later adopted that combines sequential and iterative approaches?

Solution:

1. In the initial phase, the team follows the **Waterfall model**. This model uses a strict sequential approach where each phase is completed before moving to the next, with fixed and well-defined requirements.
2. Later, the team adopts the **Spiral model**. This model combines a sequential development structure with iterative cycles, allowing regular reviews, risk analysis, and accommodation of changing requirements.